THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Temperature and its Measurement** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which temperature is not possible on a clinical thermometer scale? 1

(a) 36 °C **(b)** 42 °C **(c)** 40 °C **(d)** 20 °C

2. The difference between heat and temperature, as at this level, is that temperature tells: 1

(a) how hot something is **(b)** its colour

(c) how much it weighs **(d)** its volume

3. Which statement is correct? 1

(a) a larger object always has higher temperature

(b) a bigger object is not always hotter

(c) cold objects have no heat

(d) temperature depends on size

4. A student wraps ice in a woollen cloth and another piece in a metal foil. Which melts first? 1

(a) the one in metal foil **(b)** the one in wool

(c) both equal **(d)** neither

5. Which of the following is the correct sequence of temperatures from lowest to highest? 1

(a) body, room, ice, boiling **(b)** ice, room, body, boiling water

(c) room, ice, body, boiling **(d)** boiling, body, room, ice

6. A pan of boiling water and a bathtub of warm water: which has higher temperature? 1

(a) equal **(b)** boiling water **(c)** cannot say **(d)** bathtub

7. Which of these helps in storing ice cream for long? 1

(a) open plate **(b)** aluminium plate **(c)** iron box **(d)** thermocol box

8. A cup of tea at 90 °C loses heat quickly at first and slowly later. The reason is that: 1

(a) the cup is heavy

(b) the temperature difference with the room reduces

(c) heat is absorbed

(d) the room is cold

9. **Assertion (A):** A clinical thermometer can be used to measure the temperature of boiling water. 1

Reason (R): A clinical thermometer has a range of about 35 degrees C to 42 degrees C.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Heat flows from a hot object to a cold object. 1
Reason (R): A hot cup of tea left on a table becomes cold because it loses heat to the surroundings.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a clinical thermometer kept under the tongue or in the armpit for about a minute? 2

12. Why is a thermometer's reading checked with fixed points? 2

13. Why can the temperature of boiling water not be raised above 100 °C in an open vessel on heating continuously? 2

14. Why is it necessary to dry the lab thermometer after dipping in a hot liquid before putting in cold? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cup of tea was cooled from 80 °C to 40 °C in 10 minutes. It cooled faster in the first 5 minutes than in the next. Explain. 3

16. Why do metals feel colder than wood at 15 °C but warmer than wood at 50 °C? 3

17. Why does a metal bottle sweat on the outside when filled with cold water? 3

OR

Explain, with reasons, why we should not place a mercury thermometer in boiling water directly without checking its range. 3

18. A hot piece of iron (100 °C) is dropped into 1 L of water at 30 °C. State what happens to the temperatures. 3

OR

Explain why a layer of snow can keep the ground warm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a lab, Vikram put a laboratory thermometer into a beaker of hot water. He made sure the bulb was fully in the water but not touching the bottom or the sides, and read the thermometer with his eyes level with the mercury thread. The reading was 70 degrees C. 5
(i) Name the thermometer used.
(ii) Why should the bulb not touch the beaker?
(iii) How should the eye be placed while reading?
(iv) After a few minutes the reading falls to 55 degrees C. What does this show about the flow of heat?

20. Analyse the data: time (min) 0, 5, 10, 15 ; temperature (°C) 80, 60, 48, 42 for a cooling cup. Describe the pattern and predict the trend. 5

OR

Explain how the sense of touch judges rate of heat flow rather than temperature, using iron and wood examples. 5

END OF QUESTION PAPER — **Total: 40 Marks**